ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 1 style (theory and calculations). Section A: objective questions (30) · Section B: closed-loop control, PID action and system response, sensors and transducers, 4–20 mA signals, motor power and torque, gearboxes, hydraulic cylinder forces and flow, MTBF, MTTR, availability and system reliability (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A system that uses feedback to correct its output is a/an … system.

A open-loop B manual C closed-loop D sequential

1.1.2 In a PID controller, the integral (I) action mainly removes …

A steady-state error. B electrical noise.
C the setpoint. D the sensor signal.

1.1.3 A strain gauge measures …

A light intensity. B temperature.
C sound. D a very small change in length.

1.1.4 Torque is calculated as …

A P × t. B F × r. C V × I. D m × v.

1.1.5 A thermocouple produces a small voltage from …

A a temperature difference between two junctions. B light falling on it.
C a magnetic field. D a change in pressure.

1.1.6 A reduction gearbox with a ratio of 5 : 1 …

A increases both speed and torque. B decreases both speed and torque.
C decreases speed and increases torque. D changes nothing.

1.1.7 Pascal's law states that pressure in a confined liquid …

A decreases with distance. B acts equally in all directions.
C acts only downwards. D is always zero.

1.1.8 MTBF stands for …

A maximum torque before failure.
B motor test by frequency.
C minimum time between faults.
D mean time between failures.

1.1.9 In a control system, the error is the …

A setpoint minus the measured value.
B output multiplied by the gain.
C input plus noise.
D supply voltage.

1.1.10 An LVDT is used to measure …

A temperature.
B flow.
C linear displacement.
D colour.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 An open-loop system cannot correct itself when a disturbance occurs.

1.2.2 Too much proportional gain can make a system oscillate.

1.2.3 The derivative (D) action reacts to the total error added up over time.

1.2.4 A potentiometer can be used as a position sensor.

1.2.5 Mechanical power is torque divided by angular speed.

1.2.6 At the same pressure, a cylinder with a larger piston area produces a smaller force.

1.2.7 Availability improves when the mean time to repair (MTTR) is shorter.

1.2.8 A transducer changes one form of energy into another.

1.2.9 Hydraulic systems are used where very large forces are needed.

1.2.10 A reduction gearbox increases the output speed.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	The desired value of a controlled variable	A	torque
1.3.2	Amount by which the output goes past the setpoint	B	MTTR
1.3.3	Time for the output to stay within a small band around the setpoint	C	setpoint
1.3.4	Sensor whose resistance changes with temperature	D	disturbance
1.3.5	Device that changes one form of energy into another	E	transducer
1.3.6	Turning effect of a force	F	overshoot
1.3.7	Average running time between breakdowns	G	gear ratio
1.3.8	Average time taken to repair a fault	H	settling time
1.3.9	Input speed divided by output speed of a gear train	I	thermistor
1.3.10	Unwanted change that upsets the output of a system	J	MTBF

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: CONTROL SYSTEMS

2.1 Draw a labelled block diagram of a closed-loop speed-control system for a conveyor motor. Show the setpoint, comparator (summing point), controller, motor drive, motor and conveyor, speed sensor and feedback path. (6)

2.2 Explain the effect of EACH of the proportional (P), integral (I) and derivative (D) actions of a PID controller. (6)

2.3 The speed setpoint of a motor is 1 500 rpm. After a start, the speed rises to a peak of 1 650 rpm and then settles at 1 470 rpm. (4)

2.3.1 Calculate the percentage overshoot. (2)

2.3.2 Calculate the steady-state error. (1)

2.3.3 Which action of the controller should be increased to remove the steady-state error? (1)

2.4 Give TWO examples of open-loop systems found in a factory. (2)

[18]

QUESTION 3: SENSORS AND TRANSDUCERS

3.1 Explain how a strain-gauge load cell and a Wheatstone bridge are used to weigh a load. (4)

3.2 Choose a suitable sensor for EACH application and give a reason: (a) the temperature of an oven at 400 °C; (b) the linear position of a hydraulic ram; (c) the speed of a rotating shaft; (d) the liquid level in a closed tank. (8)

3.3 A temperature transmitter gives a 4–20 mA signal for a range of 0–200 °C. (4)

3.3.1 Calculate the temperature when the signal is 12 mA. (2)

3.3.2 Calculate the signal current at 150 °C. (2)

[16]

QUESTION 4: ACTUATORS AND DRIVES

4.1 A conveyor motor delivers a torque of 25 N·m at 1 450 rpm. Calculate its output power. (3)

4.2 The motor in 4.1 drives the conveyor through a reduction gearbox with a ratio of 20 : 1 and an efficiency of 90%. Calculate the output speed and the output torque of the gearbox. (4)

4.3 A hydraulic cylinder has a bore (piston diameter) of 80 mm and works at a pressure of 10 MPa. Calculate the force when the cylinder extends. (4)

4.4 The piston rod of the cylinder in 4.3 has a diameter of 40 mm. Calculate the force when the cylinder retracts. (3)

4.5 Give TWO advantages of hydraulic actuators and TWO advantages of electric actuators. (4)

4.6 The cylinder in 4.3 must extend 0,5 m in 5 s. Calculate the oil flow rate needed in litres per minute. (2)

[20]

QUESTION 5: RELIABILITY AND MAINTENANCE

5.1 A packing robot ran for 2 000 hours and had 4 breakdowns. The total repair time for the 4 breakdowns was 40 hours. Calculate the: (6)

5.1.1 MTBF (2)

5.1.2 MTTR (2)

5.1.3 availability (2)

5.2 Explain the difference between preventive maintenance and predictive maintenance, and give ONE example of each. (4)

5.3 A machine has three parts in series, each with a reliability of 0,95. Calculate the reliability of the whole machine. (3)

5.4 Two identical pumps, each with a reliability of 0,9, are connected in parallel so that either pump can do the job. Calculate the reliability of the system and explain why it is higher than that of one pump. (3)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER